

NORTHWEST ALABAMA, AL, USA

WMO: 723235

Lat: 34.744N

Lon: 87.600W

Elev: 540

StdP: 14.41

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13896

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.5	23.4	6.0	7.7	25.0	10.2	9.5	27.6	23.8	53.7	20.4	54.7	6.3	50	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	96.1	75.3	93.8	75.1	91.7	74.8	79.0	89.6	78.0	88.5	77.1	87.2	7.0	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.1	139.1	83.9	75.1	134.2	82.7	74.1	129.8	81.7	42.7	89.8	41.7	88.6	40.7	87.1	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	16.9	14.8	13.1	99.3	4.7	3.6	9.8	101.9	7.0	104.1	4.4	106.1	1.0	108.7
			11.6	80.8	4.5	1.2	8.4	81.6	5.8	82.3	3.2	83.0	0.0	83.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.4	42.2	46.0	53.6	62.7	70.7	77.7	80.4	79.9	74.4	63.1	52.1	44.9
	DBStd	15.73	10.95	9.77	9.24	7.75	6.43	4.40	3.64	3.96	6.01	8.29	8.69	9.74
	HDD50	850	293	176	69	6	0	0	0	0	0	6	77	223
	HDD65	2932	708	533	368	132	25	0	0	0	8	136	395	627
	CDD50	5373	51	65	180	388	642	831	943	926	732	412	140	63
	CDD65	1983	1	3	14	64	203	382	478	461	289	78	8	2
	CDH74	19087	0	9	118	572	1825	3734	4821	4532	2653	766	54	3
	CDH80	8277	0	0	15	115	651	1695	2272	2169	1141	213	6	0
Wind	WSAvg	5.7	7.1	7.3	7.5	7.2	5.8	4.7	4.0	3.7	4.3	5.0	5.7	6.6
Precipitation	PrecAvg	54.3	4.5	4.5	5.9	4.4	5.0	4.5	4.4	3.3	4.1	3.1	4.6	5.6
	PrecMax	78.8	11.8	10.7	14.3	12.8	11.9	13.4	14.4	7.2	10.1	8.3	11.1	19.2
	PrecMin	39.2	0.6	0.5	1.3	0.4	0.8	1.0	0.9	0.2	0.1	0.0	1.1	0.6
	PrecStd	10.9	2.3	2.3	3.1	2.6	2.6	2.6	2.7	1.9	2.4	2.2	2.0	3.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.5	75.2	82.3	86.5	92.1	96.9	98.4	100.3	96.6	89.1	79.9	73.6
		MCWB	62.4	63.3	64.5	67.3	72.3	74.8	75.8	75.2	72.1	69.8	66.1	64.9
	2%	DB	67.0	70.8	77.3	83.3	89.1	93.6	95.3	96.1	93.3	85.1	74.4	69.2
		MCWB	60.1	61.1	62.5	66.0	71.4	74.7	76.8	75.8	72.7	69.0	62.3	62.4
	5%	DB	63.7	66.7	73.5	80.5	86.6	91.5	93.1	93.2	90.2	81.9	70.9	65.0
		MCWB	58.5	58.2	60.2	64.8	70.1	74.2	76.4	75.2	72.7	66.9	60.9	59.6
	10%	DB	59.8	62.6	69.5	77.2	83.9	89.2	90.7	90.6	86.8	78.5	67.4	61.0
		MCWB	53.9	55.0	58.4	63.4	69.3	73.7	75.9	75.0	71.7	65.1	59.3	55.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.0	66.1	67.4	71.1	75.7	79.2	80.8	80.3	77.1	74.3	68.8	66.7
		MCDB	69.0	72.0	76.5	82.1	86.0	89.9	90.9	92.5	87.4	83.1	74.4	71.4
	2%	WB	62.0	63.8	64.9	69.0	73.9	77.9	79.3	78.5	75.7	72.4	66.1	63.7
		MCDB	65.8	69.3	73.8	78.6	84.6	88.8	89.7	89.5	86.4	81.0	71.6	68.0
	5%	WB	59.2	60.0	62.6	67.5	72.5	76.6	78.3	77.5	74.7	70.3	63.8	60.4
		MCDB	63.5	64.9	70.8	76.3	82.8	87.0	88.5	88.0	84.7	77.6	68.6	64.4
	10%	WB	55.3	56.7	60.3	65.6	71.1	75.4	77.3	76.5	73.5	68.0	60.7	56.2
		MCDB	58.4	61.4	67.9	74.2	80.6	85.0	87.4	86.3	82.6	74.8	65.9	60.4
Mean Daily Temperature Range	5% DB	MDBR	18.1	19.1	20.7	21.7	20.8	20.1	19.2	20.5	22.1	23.5	21.9	18.0
		MCDBR	20.4	23.6	25.4	24.8	23.9	23.9	22.6	24.7	25.6	27.1	26.2	21.0
		MCWBR	15.1	15.4	13.9	11.0	9.1	8.2	7.3	7.8	8.9	12.2	15.5	15.1
	5% WB	MCDBR	18.0	19.0	21.3	19.8	20.0	19.8	19.5	20.1	20.4	20.5	19.2	18.3
		MCWBR	15.1	14.6	13.5	10.2	8.7	8.4	7.6	7.8	8.2	10.8	13.7	14.9
Clear-Sky Solar Irradiance	taub	0.318	0.326	0.365	0.398	0.438	0.465	0.508	0.521	0.428	0.373	0.338	0.317	
	taud	2.494	2.470	2.386	2.288	2.223	2.171	2.080	2.031	2.282	2.416	2.457	2.514	
	Ebn at Noon	282	292	287	282	271	262	250	244	264	271	271	275	
	Edh at Noon	27	30	36	42	45	47	52	53	40	32	28	25	
All-Sky Solar Radiation	RadAvg	776	941	1306	1682	1861	1991	1901	1756	1526	1216	886	662	
	RadStd	62	118	117	113	142	154	104	98	132	160	92	65	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-2.74	N/A	N/A	N/A	N/A	N/A	+263	N/A
(47) Regional (28 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+252	+182

Nomenclature: See separate page